

Internship Report
Of
**"AICTE–EduSkills Virtual Internship: Data Science Master Program
(Siemens)"**

Submitted
To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 25SCS1003001198

Name of Student: RAJ SRIVASTAVA

Batch: 2026–27

CANDIDATE'S DECLARATION

I, Raj Srivastava, do hereby declare that the internship report titled “AICTE–EduSkills Virtual Internship: Data Science Master Program (Siemens)” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: Raj Srivastava

Roll No.: 25SCS1003001198

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their constant support and encouragement throughout this internship. I am also grateful to AICTE, the National Internship Portal, EduSkills, and Siemens Industry Software for organising and supporting this Virtual Internship Program and providing the platform, structured content, and certification that made this learning experience possible.

I express my gratitude to my parents for their blessings.

Signature: _____

Student Name: Raj Srivastava

Roll No.: 25SCS1003001198

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INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



AICTE
अखिल भारतीय तकनीकी शिक्षा परिषद
All India Council for Technical Education



NATIONAL
INTERNSHIP
PORTAL



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Raj Srivastava

IILM University, Greater Noida

has successfully completed the 8-weeks

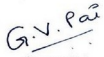
Data Science Master Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS



Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 4a9dba0a93a9770da3ae
Student ID: STU6885b67b9481b1753593467



GRADE: O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

Chapter 4: Project Description

4.1 Introduction

The rapid growth of data-driven decision-making across industries has created a strong demand for professionals skilled in data science, machine learning, and data engineering. To bridge this industry-academia skill gap, the All India Council for Technical Education (AICTE), in collaboration with the National Internship Portal and EduSkills, offers structured Virtual Internship Programs supported by leading technology companies. As part of my academic curriculum at IILM University, Greater Noida, I undertook the 8-week “Data Science Master Virtual Internship,” conducted under the AICTE–EduSkills initiative and supported by Siemens Industry Software, during June–August 2026.

This report documents the learning journey, tools, methodologies, and outcomes of the internship, which was delivered through the RapidMiner platform (an Altair company) and covered the complete data science lifecycle — from identifying business problems suitable for AI/ML, through data engineering and model building, to deployment and enterprise platform administration.

4.2 Organization Profile

The internship was jointly facilitated by the following organisations:

- AICTE (All India Council for Technical Education): The statutory body under the Ministry of Education, Government of India, responsible for planning, promotion, and coordination of technical education.
- National Internship Portal: A Government of India initiative connecting students to structured, verified internship opportunities.
- EduSkills Foundation: An ISO-certified, AICTE-recognised skilling partner that designs and delivers industry-aligned virtual internship programs across emerging technology domains.
- Siemens Industry Software (India) Pvt. Ltd.: The industry partner supporting the Data Science Master track, providing curriculum direction and real-world context for the program.

- RapidMiner (an Altair Company): The data science platform used throughout the internship for hands-on labs, assessments, and certification exams.

4.3 Problem Statement

Organizations today generate large volumes of raw, unstructured data but often lack in-house skills to convert this data into actionable business insight. The internship was structured around this real-world problem: how to take a business use case, acquire and prepare the underlying data, build and validate predictive models, and finally deploy and administer those models on an enterprise-grade platform — all in a reproducible and scalable manner.

4.4 Project Objectives

- To learn how to identify business problems that are suitable candidates for AI and machine learning solutions.
- To acquire practical skills in accessing, loading, cleaning, and transforming raw data.
- To build foundational and advanced machine learning models, including classification, regression, and clustering.
- To master advanced data preparation techniques such as scripting, macros, regular expressions, and web APIs.
- To learn how to deploy, monitor, and manage machine learning models in a production-like environment.
- To gain hands-on experience administering an enterprise AI platform (RapidMiner AI Hub).

4.5 Scope of the Project

The scope of the internship spanned the complete data science workflow taught over eight structured weeks, covering four proficiency modules: Applications & Use Cases, Data Engineering, Machine Learning, and Platform Administration — each assessed at both the Professional and Master certification levels. The scope was limited to the RapidMiner ecosystem and the datasets and case studies provided within the virtual internship curriculum, and did not extend to a live client deployment.

4.6 Technologies and Tools Used

- RapidMiner Studio — for visual workflow design, data preparation, and model building.
- RapidMiner AI Hub — for model deployment, scheduling, and enterprise administration.
- RapidMiner Radoop — for big-data processing concepts.
- Machine Learning techniques — classification, regression, clustering, association rules, cross-validation, and parameter optimization.
- Scripting and automation — loops, branches, macros, regular expressions, and Web API integration.
- R and Python integration within RapidMiner for extending model capability.

4.7 System Architecture

The internship followed a modular, layered learning architecture aligned with a typical enterprise data science pipeline:

- Data Layer: Access and ingestion of raw data from files, databases, and Web APIs.
- Processing Layer: Data cleansing, transformation, pivoting, and feature engineering within RapidMiner Studio.
- Modelling Layer: Training, validating, and optimizing machine learning models (classification, regression, time-series forecasting).
- Deployment Layer: Publishing models as web services and scoring endpoints through RapidMiner AI Hub for real-time scoring.
- Administration Layer: Managing users, marketplace extensions, and enterprise operations on RapidMiner AI Hub.

4.8 Methodology

The internship used a structured, week-wise, competency-based methodology combining self-paced e-learning, hands-on labs, and assessments:

- Week 1 – Applications & Use Cases Professional: Identifying business problems suitable for AI and ML.

- Week 2 – Data Engineering Professional: Accessing, loading, transforming, and calculating data.
- Week 3 – Machine Learning Professional: Building foundational machine learning models.
- Week 4 – Data Preparation Master: Advanced data handling and automation.
- Week 5 – Machine Learning Master: Advanced machine learning techniques.
- Week 6 – Applications & Use Cases Master: Deploying and monitoring ML models.
- Week 7 – Platform Administration Master: Configuring and managing RapidMiner AI Hub.
- Week 8 – Platform Administration Master: Enterprise-level platform operations.

Each module concluded with a proctored assessment, and the final week included an overall Final Assessment Test that, together with weekly assessments and project submissions, determined the final internship grade.

4.9 Expected Outcomes

- Successful completion of the 8-week AICTE–EduSkills Data Science Master Virtual Internship with an overall grade of B.
- Working proficiency in the complete data science lifecycle, from problem identification to model deployment.
- Seven individual RapidMiner certifications spanning Professional and Master levels across four competency areas.
- Improved employability through demonstrable, industry-recognised, verifiable digital credentials.
- A stronger foundation for applying data science techniques in future academic and personal software projects.

4.10 Certificates of Completion and Other Communication Proof

The following pages present the Internship Offer Letter and the individual RapidMiner certifications earned during the course of this internship.

Internship Offer Letter



Ref No: C17Y2026M071495450

Internship Offer Letter

Student Name : Raj Srivastava
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Siemens Data Science Master
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU6885b67b9481b1753593467

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



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Applications & Use Cases Professional Certification



This is to certify that

Raj Srivastava

successfully passed the examination for

Applications & Use Cases Professional Certification

The Professional level is the first of two certification levels and is rewarded for solid experience in the topic. Some of the topics tested at this level are:

Introduction to Machine Learning and Data Science, CRISP-DM, Use Cases for Machine Learning, Visualization and What to do with Models.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.


Knut Makowski
(Head of Education)

Data Engineering Professional Certification



This is to certify that

Raj Srivastava

successfully passed the examination for

Data Engineering Professional Certification

The Professional level is the first of two certification levels and is rewarded for solid experience in the topic. Some of the topics tested at this level are:

Data Access, Basic Transformations, Working with Multiple Data Sets, Pivot Tables, Routines and Simple Text Processing.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.


Knut Makowski
(Head of Education)

Machine Learning Professional Certification



This is to certify that

Raj Srivastava

successfully passed the examination for

Machine Learning Professional Certification

The Professional level is the first of two certification levels and is rewarded for solid experience in the topic. Some of the topics tested at this level are:

Classification, Regression, Scoring, Hold-out Validation, Correlations, Feature Importance, Clustering and Association Rules.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.


Knut Makowski
(Head of Education)

Data Engineering Master Certification



This is to certify that

Raj Srivastava

successfully passed the examination for

Data Engineering Master Certification

The Master level is the second level after Professional and stands for high proficiency in the topic. Some of the topics tested at this level are:

Loops and Branches, Advanced Text Processing, Exception Handling, Logging, Data Cleansing, Regular Expressions, Macros, Web APIs, and Scripting.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service


Knut Makowski
(Head of Education)



Machine Learning Master Certification



This is to certify that

Raj Srivastava

successfully passed the examination for

Machine Learning Master Certification

The Master level is the second level after Professional and stands for high proficiency in the topic. Some of the topics tested at this level are:

Complex Predictive Models, Cross Validation and Correct Validation, Parameter Optimization, Model Selection, Feature Engineering, Time Series and Forecasting and Integrating R and Python Models.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgespassport.com or another open badge service.


Knut Makowski
(Head of Education)

Applications & Use Cases Master Certification



This is to certify that

Raj Srivastava

successfully passed the examination for

Applications & Use Cases Master Certification

The Master level is the second level after Professional and stands for high proficiency in the topic. Some of the topics tested at this level are:

Running Processes, Deploying Models, Model Management and Web Apps.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service


Knut Makowski
(Head of Education)

Platform Administration Master Certification



This is to certify that

Raj Srivastava

successfully passed the examination for

Platform Administration Master Certification

The Master level stands for high proficiency in the topic. Some of the topics tested at this level are:

Platform Overview, RapidMiner Studio Installation, RapidMiner AI Hub Administration, RapidMiner Radoop, RapidMiner Real-Time Scoring, and RapidMiner Marketplace.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgespassport.com or another open badge service.


Knut Makowski
(Head of Education)

Chapter 5: Bibliography / References

1. AICTE (All India Council for Technical Education), National Internship Portal. <https://internship.aicte-india.org/>
2. EduSkills Foundation, AICTE–EduSkills Virtual Internship Program. <https://eduskillsfoundation.org/>
3. RapidMiner (an Altair Company), RapidMiner Studio and AI Hub Documentation. <https://docs.rapidminer.com/>
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5. Internship Offer Letter, Ref No. C17Y2026M071495450, EduSkills, 2026.
6. Certificate of Virtual Internship, Data Science Master Virtual Internship, AICTE–EduSkills, 2026.